PROJECT 5 OF 21 • 11.B — GROUP 5 • AI PROJECT (ARDUINO) – 20 MARKS

Cane Companion

Warns a blind user about an obstacle at chest height, and calls for help after a fall

BUILT IN CLASS, WITH REAL PARTS. First build and test it in Tinkercad (checkpoint marks), then assemble the real Arduino, components and wiring in the ICT lesson. **Bring this sheet to every lesson** – printed, or saved as pictures/PDF on your iPad.

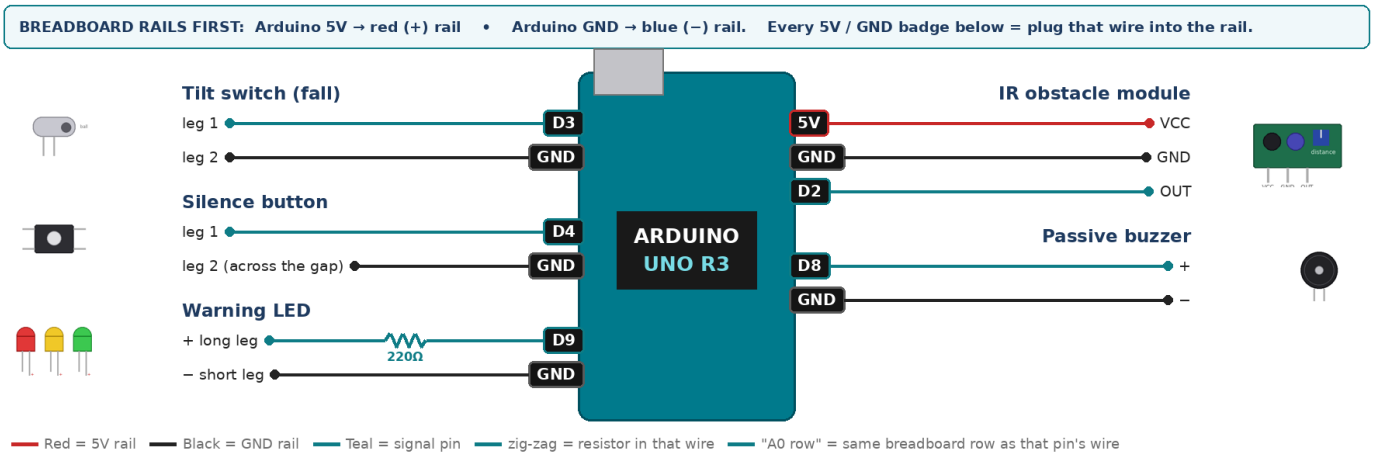
PURPOSE – WHY THIS PROJECT EXISTS

What it does	An infrared eye on the cane detects a wall, pillar or open car door ahead and gives short high beeps. A tilt switch detects that the cane has fallen flat — which usually means the user has too — and switches to a continuous low tone so people nearby come to help. A button silences it.
UAE link	People of Determination — the UAE national commitment to accessibility
Why it matters	The UAE calls people with disabilities People of Determination, and commits to making public space usable by everyone. A white cane finds obstacles at ground level but not at chest or head height — a low branch, a scaffold bar, an open van door. This add-on covers that gap for the price of a sandwich. The fall alarm is the other half: for someone living alone, the danger is not the fall, it is the hour before anybody notices.
Who it helps	People of Determination with visual impairment, and elderly people who walk with a cane.
The science	The IR module emits infrared and watches for a reflection. OUT goes LOW when something is close. It reports near or far only — never a distance in centimetres.

COMPONENTS YOU NEED

Component	Qty
Arduino UNO R3 + USB cable	1
400-point breadboard	1
IR obstacle module	1
Tilt / vibration switch	1
Push button	1
Passive buzzer	1
LED	1
220Ω resistors (red-red-brown)	1
Jumper wires (male-male)	about 20
Male-FEMALE jumper wires (for modules)	3

WIRING DIAGRAM – EVERY PIN, EVERY WIRE



WIRING TABLE – TICK EACH WIRE AS YOU PLACE IT

Component	Component pin	Goes to	Resistor	Note	✓
IR obstacle module	VCC	5V rail	—		☐
IR obstacle module	GND	GND rail	—		☐
IR obstacle module	OUT	D2	—	LOW = something close	☐
Tilt switch (fall)	leg 1	D3	—	INPUT_PULLUP	☐
Tilt switch (fall)	leg 2	GND rail	—		☐
Silence button	leg 1	D4	—	INPUT_PULLUP	☐
Silence button	leg 2 (across the gap)	GND rail	—		☐
Passive buzzer	+	D8	—	tone()	☐
Passive buzzer	–	GND rail	—		☐
Warning LED	+ long leg	D9	220Ω		☐
Warning LED	– short leg	GND rail	—		☐

BUILD ORDER – USB UNPLUGGED UNTIL STEP 7

1. Connect Arduino 5V to the red (+) rail and GND to the blue (–) rail of the breadboard.
2. Connect the IR module alone and print digitalRead(D2). Set the screw to trigger at about 20 cm.
3. Add the tilt switch; check it reads LOW when the board lies flat.
4. Add the buzzer and LED, then write the two sound patterns.
5. Tape the build to a stick so it looks like a real cane.
6. Check every wire against the table with a partner, THEN plug in the USB. The green ON light must come on – if not, unplug at once (short circuit).
7. Arduino IDE: Tools › Board › Arduino Uno, then Tools › Port.
8. Paste the code below, press Upload, open Tools › Serial Monitor at 9600 baud and watch your own readings.

STEP 1 – TINKERCAD FIRST (CHECKPOINT MARKS)

Go to www.tinkercad.com › Circuits › Create new Circuit. Build the same wiring, paste the same code, press Start Simulation. Parts that Tinkercad does not have:

- IR obstacle module → use a Pushbutton (pressed = LOW = 'something close'), same pin.
- Passive buzzer → use a Piezo (same code, tone() works).
- Tilt switch → use Tinkercad's Tilt Sensor, or a Pushbutton/Slideswitch to fake a tilt.

Save it with the name **11.B G5 – Cane Companion** and share the link with your teacher. The REAL build in class is your AI Project.

HOW THE CODE THINKS

- silence button flips the silent flag
- IR LOW means obstacle; tilt LOW means fallen
- LED on if either is true
- fallen → continuous 900 Hz tone (highest priority)
- obstacle only → short 2000 Hz chirps

THE COMPLETE CODE (TESTED)

Yellow lines = numbers YOU must change after measuring with your own sensor. Every member must be able to explain at least one line.

```
/* PROJECT 5 - CANE COMPANION  (11.B Group 5)
   IR sensor warns of an obstacle ahead (short high beeps).
   Tilt switch detects the cane lying on the floor for 1 s (long low tone).
   SILENCE button turns the sound off/on; the LED still works.          */
const int IR = 2;                       // IR OUT (LOW = obstacle)
const int TILT = 3;                     // tilt switch (LOW = cane fallen)
const int SILENCE = 4;                  // button (other leg to GND)
const int BUZZ = 8;                     // PASSIVE buzzer (tone)
const int LED = 9;                      // LED through 220 ohm

bool silent = false;
int  lastBtn = HIGH;
unsigned long tiltSince = 0, lastBeep = 0, lastPrint = 0;

// True ONCE at the moment a button goes from up (HIGH) to pressed (LOW).
// Holding the button down does NOT repeat it.
bool justPressed(int pin, int &lastState) {
  int now = digitalRead(pin);
  bool pressed = (lastState == HIGH && now == LOW);
  lastState = now;
  if (pressed) delay(25);               // ignore contact bounce
  return pressed;
}

void setup() {
  Serial.begin(9600);
  pinMode(IR, INPUT);
  pinMode(TILT, INPUT_PULLUP);
  pinMode(SILENCE, INPUT_PULLUP);
  pinMode(LED, OUTPUT);
}

void loop() {
  if (justPressed(SILENCE, lastBtn)) {
    silent = !silent;
    Serial.println(silent ? "SILENT" : "SOUND ON");
  }

  bool obstacle = (digitalRead(IR) == LOW);

  // fallen only if the tilt switch stays closed for 1 second (no false alarms)
  if (digitalRead(TILT) == LOW) { if (tiltSince == 0) tiltSince = millis(); }
  else tiltSince = 0;
  bool fallen = (tiltSince != 0 && millis() - tiltSince > 1000);

  digitalWrite(LED, obstacle || fallen);

  if (silent) {
    noTone(BUZZ);
  } else if (fallen) {
    tone(BUZZ, 900);                    // continuous: person may be down
  } else if (obstacle) {
    if (millis() - lastBeep >= 250) { tone(BUZZ, 2000, 80); lastBeep = millis(); }
  } else {
    noTone(BUZZ);
  }

  if (millis() - lastPrint >= 300) {
    lastPrint = millis();
    Serial.print("obstacle="); Serial.print(obstacle);
    Serial.print("  fallen="); Serial.println(fallen);
  }
}
```

TEST IT AND RECORD YOUR READINGS

- Walk the cane at a wall and measure the trigger distance five times. Take the average.
- Test against a white wall, a black bag and a glass door — record which one it fails on.
- Lay the cane on a cushion; the continuous tone must start within a second.

Test #	What we did	Reading / value	What happened
1			
2			
3			
4			
5			

IF IT DOES NOT WORK

- Triggers constantly: screw turned too far, or bright sunlight. Test indoors.
- Misses dark and glass surfaces — a genuine limit of infrared. Admitting it scores better than hiding it.
- Buzzer only clicks: that is the ACTIVE buzzer. tone() needs the PASSIVE one.
- Button does nothing: it is not straddling the centre gap, or both wires are on legs that are already joined. Use legs on OPPOSITE sides of the gap.

AT THE STEM WEEK TABLE

Blindfold a visitor, hand them the cane and let them walk three steps at a cardboard wall. Nothing else on the table will land as hard.

> **CHALLENGE:** Delay the fall alarm by 5 seconds and cancel it if the cane is lifted — a real device must not scream every time it is put down.

BUILD A COVER FOR YOUR PROJECT (NOTE)

Make a cover or case that suits your project design, using cardboard or materials from stationery (card, coloured paper, tape, glue, markers). **Idea:** a cane from a long cardboard tube, sensor at the tip and the buzzer near the handle.

Keep the USB port, the sensor and any buttons or knobs easy to reach. Never cover the Arduino so tightly that wires are pulled.

PROJECT POSTER (SUPPORT – 4 MARKS)

Make one clear poster (A3 paper or Canva) for the STEM Week table. It must show:

- Project name and the school, your names, class and group.
- The problem in the UAE and who it helps.
- How it works in 3 steps: sensor → Arduino decides → output.
- A photo of your real wiring and of the finished project with its cover.
- Your test results (a small table or chart) and one improvement you made.

AI LOG – PROMPTS, EXAMPLES AND DATA (CORE – 4 MARKS)

Use an AI tool (ChatGPT, Claude, Gemini or Z.ai) to help you understand and improve your project, and record EVERY prompt you use. Example prompts:

- Explain line by line what the loop() does in this Arduino sketch for a "Cane Companion": [paste the code]
- My sensor gives ___ in one condition and ___ in the other. Which numbers should I put in the CALIBRATE lines and why?
- Suggest one improvement for our "Cane Companion" that we can test in one lesson.

AI tool	Prompt we typed	What the AI answered (short)	What we used / changed

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PROJECT VIDEO – EXPLAIN THE AI PROCESS AND RESULTS (4 MARKS)

Every group – or each student – records a 1–2 minute video in class. Include:

- Who you are (names, class, group) and the problem your project solves in the UAE.
- How it works: show the real wiring and explain one sensor and one part of the code.
- The AI process: which AI tool you used, one prompt you typed and what you changed because of it.
- Your results: a live demo and your test readings – before and after your improvement.
- One thing you learned.

HOW YOUR PROJECT IS MARKED – 20 MARKS

Criterion	What you must do	For your Arduino project	Marks
Support	Choose an idea and create a clear project poster.	Your project is chosen for you – make a clear A3 / Canva poster: name, problem, how it works, wiring photo, team names.	4
Core	Use an AI tool and document the required prompts, examples, or data.	Use an AI tool (ChatGPT, Claude, Gemini, Z.ai) to understand, test or improve your code and plan your calibration. Copy your prompts and your sensor data into the AI LOG.	4
Advanced	Build, test, and improve the project using evidence.	Build the real circuit, test it, record readings, improve it – and show before / after evidence.	4
Video	Explain the AI process and results in a short video.	1–2 minute video: how it works, how AI helped you (your prompts) and what your test results show.	4
On time	Submit the complete project on time.	Poster, AI log, readings table, video and the working project – all ready by Friday 30 October.	4
TOTAL			20

Deadline: submit the complete project – poster, AI log, readings, video and working build – by Friday 30 October (end of Week 9).